



# ABDUL MUHYI BIN FAIDZULI

## ROBOTICS MAJOR

### CONTACT

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- <https://kurumin03.github.io>

### LANGUAGES

- Malay (Native)
- English (Fluent)
- Japanese (Fluent)

### COMPUTER SKILL

- Microsoft Office
- Canva,
- C language
- python
- C++
- Web designing
- Photo and video editing
- Photoshop
- Matlab
- Fusion360
- Autocad
- Solid Edge

### INSTRUMENTAL SKILL

- Lathe Machine
- Milling Machine
- Drilling Machine

### CERTIFICATIONS

- N1 Japanese Language Proficiency Test (JLPT)
- Fundamental Information Technology Engineer(FE)
- TOEIC (965 score),

### EDUCATION

- Ritsumeikan University Biwako Kusatsu Campus** APRIL 2024 - MARCH 2026  
Degree In Robotics
  - CGPA: 3.58
  - Coursework: Actuators, Applied Mathematics, Process Engineering, Statistics, Robot Vision, Measurement Engineering, Numerical Computation, Sensor, Science for Intelligence, Electrical and Electronic Circuits, Computer Programming, Exercises in Robot Design, Experiment in robot creation
- University Kuala Lumpur Malaysia France Institute** JULY 2021 - FEBRUARY 2024  
Diploma In Mechanical Engineering
  - CGPA: 3.58 (Dean's List 4 time)
  - Coursework: Mathematics, Physics, Chemistry, Control Theory, Mechanical Instrument, Thermodynamics, Information Processing, Mechanical Vibration, Fluid Mechanics, Mechanics of Materials, Machine Design, Engineering Ethics, Mechanics and Machine Elements

### WORK EXPERIENCE

- NICHIAS FGS SDN. BHD.** AUGUST 2022 - OCTOBER 2022  
Intern
  - Played a key role in supporting the technical and maintenance team by actively participating in repair and maintenance tasks.
  - Implemented new safety rules for gas meters within the production process, contributing to improved safety measures.

### PROJECT

- Path Planning using A\* Algorithm**
  - Developed an A\*-based joint-space planner for a 3-link robotic arm, integrating forward kinematics, Jacobian manipulability metrics, and heuristic cost functions to generate smooth, dexterous motion trajectories.
  - Built a full simulation pipeline with trajectory visualization, manipulability ellipsoid analysis, and comparative evaluation against Dijkstra's algorithm, achieving significantly faster optimal path generation.
- Ping Pong Robot**
  - Built an autonomous ping-pong robot with real-time ball tracking using computer vision, trajectory prediction, and a hybrid PC-Arduino control system to accurately position a two-axis paddle mechanism.
  - Designed and integrated mechanical, electronic, and control systems and demonstrating full system functionality in live rally tests.
- Development of Safety Cover at Horizontal Milling Machine**
  - Created a full safety-cover system for a horizontal milling machine, integrating CAD modeling, and functional testing to increase safety in workplace
  - Conducted hazard analysis, evaluated guard performance through experiments, and delivered a validated protective design that improved machine safety without affecting workflow.

### RESEARCH

- Humanoid Systems Laboratory** APRIL 2024 - MARCH 2026  
AUTOMATED DOCKING APPROACH PHASE FOR UR5E ROBOT USING YOLOV8
  - Developed an automated docking approach system for a UR5e robot using YOLOv8, TF2, and MoveIt, enabling end-to-end detection-to-motion execution and achieving reliable autonomous alignment in simulation.
  - Designed and implemented a full perception-planning-control pipeline, including 3D pose estimation, IK-based trajectory generation, and safety-oriented motion execution; currently extending the system for deployment on the real UR5e robot.